Chapter 1

Eulerian Hydrodynamics

1.1 Introduction

The topic of gas dynamics is extensively covered in our course AS7002, "Astrophysical Gas Dynamics", which is a prerequisite for this course. The reason gas dynamics is so important for a course on computational astrophysics is that we learn the equations of motion that describe gases and fluids in the universe. Even in the first project you saw some of these equations in the smooth particle hydrodynamics section. In this project we will revisit the basics of gas dynamics as an introduction before discussion the numerical methods we use to solve the equations. Even after a full course like AS7002, practical numerical exercises of gas dynamics would be a tall order, so the exercises we do here will be more fundamental than astrophysical. However, I hope this gives a taste of what the field is like and you learn somethings about numerical hydrodynamics.

There are many great resources for the topic of gas dynamics and their numerical solutions. This include "Principles of Astrophysical Fluid Dynamics" by C.J. Clarke and R. F. Carswell (Clarke & Carswell 2014), "Riemann Solvers and Numerical Methods for Fluid Dynamics: A Practical Introduction" by E. F. Toro (Toro 2009), "Introduction to Computational Astrophysical Hydrodynamics" by Michael Zingale & the Open Astrophysics Bookshelf (Michael Zingale 2025).

1.1.1 Gas Dynamics in Astrophysics

Most of the material we encounter in the Universe can be describe as a fluid or a gas, we'll use these terms interchangeably. Many of the dynamical processes (and even many of the static ones) we observe in the Universe can be understood in term of the physics of fluids. The problem of treating astrophysics fluids is a bit different from terrestrial fluid mechanisms/dynamics. In an astrophysics context we typically have no solids or obstructions and we generally have smooth flows (although shocks are definitely allowed). Unlike terrestrial fluid mechanisms, our fluids are typically *compressible*, this changes the equations a bit. Often we ignore viscosity since the so-called kinematic viscosity is very, very small. Viscous-like effects often show up, for example in accretion disks, but typically they have an origin other than a direct viscosity, i.e. magnetic fields in accretion disks. Sometimes we can model these effects with an effective viscosity.

Finally, while we say often say fluid *dynamics*, we don't only study fluids and gases that move. In general, we study fluids and gases that are responding the external (or internal) forces: gravity, pressure gradients, or radiation forces. Often this can give static structures, like stars, but this is still part of astrophysical gas dynamics.

Some examples of astrophysics systems that we can work with are stars, white dwarfs, neutron stars, interstellar medium, intergalactic medium, supernovae, stellar winds, galactic winds, molecular clouds, ionized hydrogen regions, jets, accretion disks (on all scales from protoplanetary disks to AGN), gas giants, and many more.... In reality, of course, these systems are composed of a large number of particles, atoms, ions, molecules, be it hydrogen, helium, or some other composition. In many scenarios in astrophysics we can treat these particles statistically and work in the continuum limit. This project introduces and leads you through solving the continuum equations of fluid dynamics. A part this statistical treatment is an understanding of the internal microphysics of the gases. This is often summarized in the equation of state, some you might be familiar with are the ideal gas equation of state, degenerate electron gases, nuclear equation of state, radiation equation of state.

1.2 Fluid Elements

As an introduction to the equations of fluid dynamics we will introduce the concept an a *fluid element*. This concept is important for understanding and working with the theory of fluid dynamics, but it is also a critical concept for numerical fluid dynamics. We show a cartoon depiction of a fluid element in Figure 1.1. Fluid elements have some basic properties.

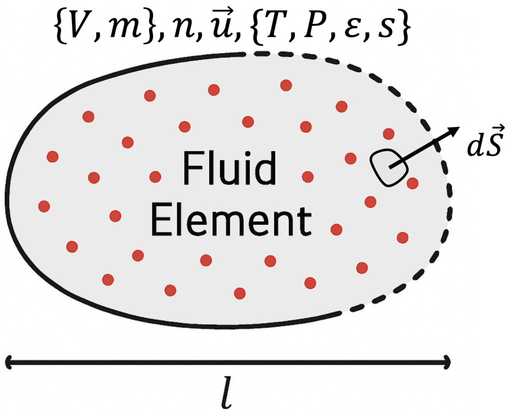

Figure 1.1: Depiction of a fluid element. Fluid elements can have either a fixed mass (and therefore a impermeable but flexible boundary) or fixed volume (and therefore a permeable but rigid boundary), a number density, and then internal properties, like temperature, pressure, internal energy, or entropy. These latter four are often related through the equation of state. The surface is denoted by the surface element $d\vec{S}$.

1. There must be a large amount of individual particles in each element so that we can treat the collection of them statistically: $V \times n \gg 1$ where n is the number density of particles and V is the volume of the fluid element.
2. The macroscopic properties of the fluid, like the density, temperature, etc, must vary over lengths scales much larger than the size of the fluid element: $q/\nabla q \gg l$, where q represents the properties of the fluid. This means that *within the fluid element*, the fluid properties are uniform.
3. The length scale of particle interactions λ, e.g. the mean free path, must be small compared to the length scale of the fluid elements: $l \gg \lambda$. This is to ensure the particles remain in local equilibrium with the particles in the system and their properties follow the expect local distributions, i.e. the particle are collisional. If they particles did not interact on length scales smaller than the fluid element than the properties of the particles could be non-local (i.e. they would depend on what is happening in other fluid elements).

1.2.1 Equation of State

When we say the particles follow a statistical distribution we mean that on a microscopic level we know the distribution of particle properties. For example, for a non-interacting (but still interacting enough to satisfy #3 above) ideal gas at temperature T, the distribution of speeds is given by the Maxwell-Boltzmann distribution,

$$n(v)dv = n\left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp^{-mv^2/k_BT} 4\pi v^2 dv \tag{1.1}$$

where $n = \int n(v)dv$ is the total number density, T is the temperature, and m is the mass of the particle. Using this distribution we can express the macroscopic properties of the fluid elements as functions of each other. For example, the energy per unit mass: $\epsilon_{\text{int}} = 3k_BT/2m$, the pressure: $P = nk_BT$.

Often we use polytropic equations of state, i.e. an equation for the pressure that only depends on the density. Then, $P = K\rho^\gamma$. It follows that $P = \rho\epsilon(\gamma - 1)$ and $c_s^2 = \gamma P/\rho$. This is the equation of state we will use in this project. The ideal gas equation of state can be reformulated as a polytrope if we make the assumption that the entropy stays constant, i.e. an adiabatic equation of state. For the ideal gas, $\gamma = 5/3$.

1.2.2 Eulerian vs. Lagrangian

We can conceptually think of two different types of fluid elements. If we imagine the fluid element to encompass a fixed set of particles so that particles cannot enter or leave the fluid element and the boundary of the fluid element to be flexible so that it moves with the bulk fluid, then we say the fluid element is a Lagrangian fluid element. The mass (or number of particles) of the fluid element is fixed since no particles can enter or leave, the volume is allowed to change, so the density changes as well. The other picture of fluid elements is the Eulerian description. Here, the volume (and often the spatial location) of the fluid element is fixed but the fluid can enter and leave the fluid elements by flowing through the boundaries. Each description of comes with its own pros and cons and sets of equations. Often the Lagrangian description is useful. For example in stellar evolution we use the Lagrangian view and derive the equations of stellar structure in terms of the mass coordinate: e.g. $dP/dm = -Gm/4\pi r^4$ or $dr/dm = 1/(4\pi r^2 \rho)$. For doing multidimensional fluid dynamics the Lagrangian picture is more difficult, although the smoothed particle hydrodynamics project you did in the first project is such an example. The Eulerian descriptions formulates the equations in terms of the spatial coordinates rather than mass coordinates: e.g. $dP/dr = -G\rho m/r^2$ or $dm/dr = 4\pi r^2 \rho$. The Eulerian description is what we will use in this project. This means we will set up fluids and track the density, energy, velocity, pressure, etc., at various fixed locations.

1.3 The Euler Equations

1.3.1 Continuity Equation

Consider the fluid element from Figure 1.1 in the Eulerian description. The boundaries are fixed in space. *How does the mass of the fluid element change in time?* The mass of the fluid element is just $\int_V \rho dV$, and therefore the change in the mass is just $\partial_t \int_V \rho dV$. But we also can describe the change in the mass of the element due to the movement of matter across the surface. The mass flow across the surface element $d\vec{S}$ and out of the element is $\rho\vec{u}\cdot d\vec{S}$, so the total change in mass is $-\int_S \rho\vec{u}\cdot d\vec{S}$. Using the divergence theorem, $-\int_S \rho\vec{u}\cdot d\vec{S} = -\int_V \vec{\nabla}\cdot(\rho\vec{u})dV$. Since mass entering or exiting through the surface is the only way for the mass to change, our first equation for fluid dynamics is given by

$$\frac{\partial}{\partial t}\int_V \rho dV + \int_V \vec{\nabla}\cdot(\rho\vec{u})dV = 0\,, \tag{1.2}$$

or, since this must be true for all volumes,

$$\boxed{\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot(\rho\vec{u}) = 0; \qquad \frac{\partial \rho}{\partial t} + \partial_i(\rho u_i) = 0}\,. \tag{1.3}$$

This equation is called the continuity equation and at its core represents the idea of mass conservation. Note the equation on the right is simply the index notation version of the continuity equation in Cartesian coordinates; $\vec{\nabla}\cdot\vec{u} = \nabla_i u_i = \partial_i u_i = \partial_x u_x + \partial_y u_y + \partial_z u_z$. For simplicity we'll restrict ourselves to Cartesian coordinates in this project[1].

1.3.2 Momentum Equation

We can do a similar derivation for the momentum equation. However we now have to consider the impact of forces. Unlike for the mass equation, forces in the system can generate momentum. The two sources we will deal with are gravity and pressure. Gravity is conceptually easy. An external gravitational field ($\vec{g} = -\vec{\nabla}\phi$; where ϕ is the gravitational potential) will accelerate the whole fluid element and impart momentum via $\int_V \rho\vec{g}dV$. The pressure is a bit more tricky.

[1] In the PhD level exercises at the end of this chapter we make to the move to other coordinate systems

Interior to the fluid element the pressure is constant and therefore on any random plane *interior* to the element the pressure applies a force (per unit area; thats what pressure is) equally to both sides of the plane. However, at the surface of the element the pressure is only acting outward as there is no part of our fluid element exterior to the element (by definition). For this reason there is a net force on the surface from the pressure interior that wants to expand the element. Of course the neighboring element also want to do the same, but its pressure might be slightly different. The force in some direction $\hat{n}$ on the element from the pressure interior is $\int_S P\hat{n} \cdot d\vec{S}$. Since the element is doing this work on the environment, this results in a loss of momentum in the $\hat{n}$ direction, so the source term (again using the divergence theorem) is $-\int_V \vec{\nabla} \cdot (P\vec{n})dV$. Like the continuity equation, momentum can also enter/exit the element because the fluid is moving. We'll leave the full derivation for AS7002 since there are some subtleties in it, but the conservative form of the momentum equation is given as,

$$\boxed{\partial_t(\rho u_j) + \partial_i(\rho u_i u_j + \delta_{ij} P) = \rho g_j} \,. \tag{1.4}$$

Here we only use index notation since it is otherwise ambiguous what quantities the divergence is taken over. For clarity, in one dimension,

$$\frac{\partial \rho u}{\partial t} + \frac{\partial}{\partial x}(\rho u^2 + P) = \rho g \,, \tag{1.5}$$

and in two dimensions there are two momentum equations, one for ρu_x,

$$\frac{\partial \rho u_x}{\partial t} + \frac{\partial}{\partial x}(\rho u_x^2 + P) + \frac{\partial}{\partial y}(\rho u_x u_y) = \rho g_x \,, \tag{1.6}$$

and one for ρu_y,

$$\frac{\partial \rho u_y}{\partial t} + \frac{\partial}{\partial x}(\rho u_y u_x) + \frac{\partial}{\partial y}(\rho u_y^2 + P) = \rho g_y \,. \tag{1.7}$$

Just some notes on these equations to build intuition: The first spatial derivative term, $\partial_x \cdots$ represents the flux in the x direction of the j-momentum. So if there is a gradient in the x direction of the j-momentum (i.e. ρu_j), there will be a net flux of j-momentum ($\sim \rho u_j u_x$) through the boundaries of the element (i.e. a different amount entering the cell through one x interface than is leaving though the other x interface). The second term is similar but for the y direction. The pressure gradient term which has a δ_{ij} only imparts momentum to the matter in the direction of the gradient. The pressure term is sometimes considered as a source term on the other side of the equation, but not here.

1.3.3 Energy Equation

The final equation is one for the energy. The energy of the fluid element has several contributions. First, the internal energy of the fluid itself is energy. We gave an expression for this in Section 1.2.1. This energy accounts for the thermal motions of the particles on a microscopic level, which are moving even if the bulk fluid is not. Internal energy can also come from degeneracy/quantum effects. Another contribution to the energy is the bulk kinetic energy of the fluid. Finally, gravitational potential energy is a form of energy we must account for. Fluid processes can transform energy between internal energy, bulk kinetic, and gravitational energy, but since energy is conserved, it useful to have an equation for the total energy evolution. The energy equation, in conservative form is given by,

$$\boxed{\frac{\partial E}{\partial t} + \vec{\nabla} \cdot [(E+P)\vec{u}] = 0; \qquad \frac{\partial E}{\partial t} + \partial_i[(E+P)u_i] = 0} \tag{1.8}$$

where $E = \rho(u^2/2 + \epsilon_{\text{int}} + \phi)$, $u^2 = |\vec{u}|^2$, and ϕ is the gravitational potential. Note for a polytropic equation of state, $\rho\epsilon_{\text{int}} + P = P/(\gamma-1) + P = \gamma P/(\gamma-1) = \rho c_s^2/(\gamma-1)$, so often the flux is expressed as $E + P = \rho[u^2/2 + \phi + c_s^2/(\gamma-1)]$.

1.3.4 The Euler equations in compact form

We can conveniently write these equations in a compact form

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F_i}(\mathbf{U})}{\partial x_i} = \mathbf{S}(\mathbf{U})\,, \tag{1.9}$$

where $\mathbf{U}, \mathbf{F}_i(\mathbf{U})$ and $\mathbf{S}$ are lists of variables representing the equations presented here, specifically $\mathbf{U} = [\rho, \rho u_j, E]$, $\mathbf{F}_i(\mathbf{U}) = [\rho u_i, \rho u_i u_j + \delta_{ij} P, (E+P)u_i]$, and $\mathbf{S} = [0, \rho g_j, 0]$. $\mathbf{U}$ are called the conserved variables. Since some of our physics (like the equation of state) depends not on the conserved variables directly we need to recover the so-called primitive variables ($\mathbf{p} = [\rho, u_j, P, \epsilon, \phi]$) after the Equations 1.9 are evolved. In general $\mathbf{U} = f(\mathbf{p})$ and $\mathbf{p} = g(\mathbf{U})$. The latter function g is not always easy to determine and sometimes it must be solved numerically. Just to be clear, this expression in Eq. 1.9 is only valid for Cartesian coordinate since there the divergence is simply $\vec{\nabla} \cdot \mathbf{F} = \partial_i F_i = \partial_x F_x + \partial_y F_y + \partial_z F_z$.

1.4 Numerically solving differential equations in space and time

In a few rare simple cases we can use the equations presented above to completely solve astrophysical problems. Rather, usually the equations help us understand what to expect in simplified systems or how perturbations will evolve and solutions require numerical methods. This is mainly due to the nonlinearity of the Euler equations. Even this numerical aspect is difficult due to the incredibly complex nature of the Euler equations (even ignoring things like viscosity like we have done here) and the computational resources needed to simulate them.

So, lets start with something a little easier, in one dimension, then use what we learn to eventually take a stab at the Euler equations in multiple dimensions in the main project.

1.4.1 The Advection Equation

Let's take the continuity equation and expand the divergence term, $\partial_t \rho + \rho \partial_x u + u \partial_x \rho = 0$. If we take u to be constant, then this further simplifies to $\partial_t \rho + u \partial_x \rho = 0$. This is the advection equation. The formal solution is just $\rho(x,t) = \rho_0(x - ut, t = 0)$ where ρ_0 is the initial profile at t=0.

To numerical solve this, we first have to discretize the advection equation. The discretization occurs in both space and time. So instead of knowing the density for all values of x, we can only know it for a set of discrete values x_m. We are ultimately interested in the time evolution of the density, but in practice the solution is also discretized in time, so we can know the solution for a series of times t_n. It is good to introduce some notation to remain consistent through the notes, see Figure 1.2

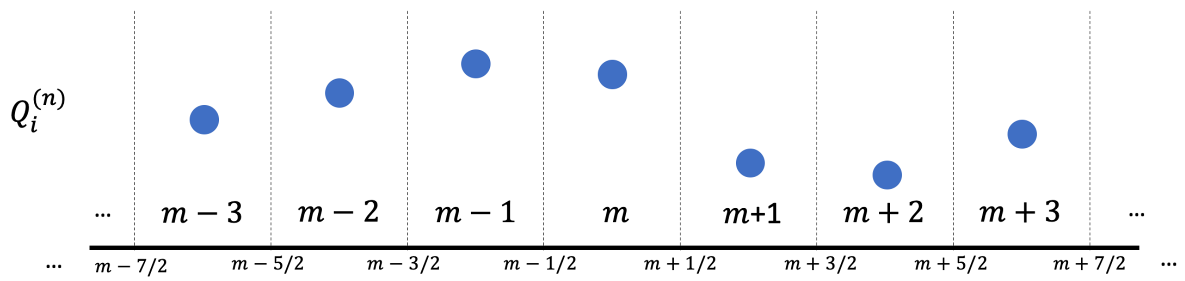

Figure 1.2: Simplistic notation for our numerical grid in one dimension

Here we have 7 zones, with each zone labeled by an integer $m-3, ..., m, ..., m+3$ at time t_n. We label the interfaces between the zones with half integer values. Each zone has a specific value of Q, but we can think of this as the density, $\rho_m^{(n)}$ marked with the blue dot. Since we are discretizing the domain, we don't have any more information than this. The finite-volume method of numerically solving the equations takes this value to be the cell average value, i.e. $\rho_m \equiv \overline{\rho} = \int_{V_m} \rho dV / \int_{V_m} dV$. (Think of this as the volume of the fluid element we had back when deriving the equations in Section 1.3). This shows the state of the system at some time t_n. We want to solve for the state of the system at some future time, say $t_{n+1} = t_n + \Delta t$, i.e. $\rho_m^{(n+1)}$.

Now, with this picture in mind, let us write down a[2] discretized version of the advection equation,

$$\frac{\rho_m^{(n+1)} - \rho_m^{(n)}}{\Delta t} + \frac{u\rho_{m+1/2}^{(n)} - u\rho_{m-1/2}^{(n)}}{\Delta x} = 0\,. \tag{1.10}$$

Note, u is constant in this toy problem, so we do not have to worry about discretizing it, in general we do and this, in part, is what makes the hydrodynamic equations non linear. First, I'll just point out that this form ensures conservation of the mass. The amount of density added to one zone is subtracted from the neighboring one. But what do we take for $\rho_{m\pm1/2}^{(n)}$? This question is known as the **Riemann problem**. For the more general case of Equation 1.9, we need to know $[\mathbf{F}_i(\mathbf{U})]_{m\pm1/2}$. We'll revisit this below and for now focus on our simple advection equation case. Taking the average of the two zones on either side of the interface $m + 1/2$, a method called "centered difference" turns out to be a bad choice (try for yourself in the exercise below). A better option is the *upwinding* method. This method looks at the value of the velocity and determines the value at the interface based on the direction of the fluid flow. If the flow is positive ($u > 0$), in our case moving from zone m to zone $m + 1$, then take $\rho_{m+1/2}^{(n)} = \rho_m^{(n)}$. If it is negative ($u < 0$), in our case moving from zone $m + 1$ to zone m, then take $\rho_{m+1/2}^{(n)} = \rho_{m+1}^{(n)}$. (Aside: More generally we can upwind even if u isn't constant, then we check both u_m and u_{m+1}. If their sign is the same, we use that sign. If they are different, we take $u_{m+1/2} = 0$.)

1.4.2 Choosing a time step

The last bit of theory we need to cover before we look at a numerical implementation of the advection equation is how to choose the time step Δt in Equation 1.10. To see why this is important consider (a) how much mass is in zone m; $m_m = \rho_m \Delta x dA$, (b) the rate of mass advecting through the interface $m + 1/2$ if it was traveling with speed u; $\dot{m}_m = \rho_m u dA$ and (c) how long it would take for all the mass to advect through the interface $t = m_m/\dot{m}_m = \Delta x/u$. This is strong limit on how big our time step can be. We should not have a time step that moves a significant amount of material (compared to what is in the zone itself) through the interface. Even though mass will be coming through the other boundary to replenish the contents, that may have a different density or be coming in faster or slower. In general, we restrict the time step to be a fraction of the 'crossing time' derived above,

$$\Delta t = C\Delta x/u\,. \tag{1.11}$$

A derivation can show that the upwinding scheme we presented here is stable if $C < 1$[3]. In practice, consider values of C between 0.25 and 0.5. In summary, the faster the matter is moving the smaller the time step must be. Furthermore, the smaller the spatial resolution, the smaller the time step must be. This is a real killer in numerical simulations of hydrodynamics. If you want to double the resolution, and study the same initial conditions for the same period of time

[2] This is just one of many ways to write down a discretized version. Numerical solutions to differential equations is a whole field of research and I'm leading you in the direction of a basic, yet useful and robust scheme for a numerical solution.

[3] The overall theory of time stepping within different numerical schemes has a large theoretical backing. Stability of different schemes are often written via the CFL (Courant–Friedrichs–Lewy) condition. This condition says that $C = u\Delta t/\Delta x \leq C_{\max}$.

in 3D this results in 2^3 times more computational fluid elements and 2 times more time steps, therefore the problem is 16 times as computational expensive.

One more note on this that we will return to in the next section. For the hydrodynamic equations in Equation 1.9, we do not have strict advection problems. In fact the equations permit sound waves to travel, this means that the relevant speed is not the fluid velocity but rather the fluid sound speed, which for subsonic problems can be significantly faster than the fluid velocity, further restricting the time step.

1.4.3 Exercise 1:

Use the template Jupyter notebook I have provided to develop an upwinding solver for the linear advection equation. Use a domain extending from $x = 0$ to $x = 1$ with a varying number of computational zones, $N = 50, 100, 200, 400, 800$. For the initial density, take a Gaussian pulse, $\rho = \exp^{-((x-x_0)/\sigma_x)^2}$ with $x_0 = 0.5$ and $\sigma_x = 0.2$. For the velocity take $u = 1$ everywhere. Use periodic boundary conditions so that as the matter leaves the domain at $x = 1$ it reenters at $x = 0$. Evolve the system for one time unit. Since $u = 1$, this should be one complete cycle of the pulse and it should be re-centered on the domain at $x = x_0$.

Do this for each N and for each one measure the norm (Euclidean norm, or sometimes called the L_2 norm) of the error. Since we know the solution should just be the initial data, this quantity is simply $\epsilon = \sqrt{\Sigma_m(\rho_m - \rho_m^0)^2/N}$ and represents some measure of the average error in the solution. The rate at which the error decreases as N increases is the order of the method. You should see that this upwinding scheme we have been using is first order, that means $\epsilon \propto 1/N^1$ (ideally (which unfortunately not always the case) in a second order method the error would go like $1/N^2$).

1.5 Second order methods

First order schemes are not ideal, and we would like to have higher order so that the error drops quickly as we increase the resolution (remember the computational cost in 3D scales like $(1/\Delta x)^4$). In this section we look at two ways to increase the order of the methods we use. In fact there are two sources of first order convergence in the exercise above in Section 1.4.3, one due to the way we evolve the system in time and the other due to how we compute the interface values. To move beyond first order we must change both of these aspects of our solution.

We will do this within the *Method of lines* formalism. The idea is to separate the calculation of the spatial flux terms from the time evolution and handle the latter with a Runge-Kutta like scheme. Other methods, either implicit methods (where you solve for the entire $(n + 1)$ state of all the variables at the same time) or characteristic tracing (where one calculates the fluxes at a intermediate time between $t^{(n)}$ and $t^{(n+1)}$), exist but we will not discuss them here.

1.5.1 Second order in time

We can extend the order of the time integration by using Runge-Kutta (RK) methods. Briefly, RK methods advance the solution some fraction of the time step. We will use the midpoint method, which is a very simple two stage RK method, but the extension to higher order methods is very programmatic. This is depicted in Figure 1.3. The two stages proceed as follows. 1. You calculate/evaluate the derivative of the function (i.e. the differential equation; in our case this includes the spatial flux term and any source terms) at time t^0; $dy/dt|^{(0)}$ (this is the blue slope in Figure 1.3). 2. You advance the unknown quantity by half of the intended time step, to $y^{(1/2)}$ using the evaluated derivative from $t^{(0)}$ (this is the blue dot in Figure 1.3). 3. You calculate a new derivative based on the solution at $t^{(1/2)}$; $dy/dt|^{(1/2)}$ (this is the orange slope in Figure 1.3). 4. You advance the unknown quantity a full time step from $t^{(0)}$ to $t^{(1)}$ using the evaluated derivative from $t^{(1/2)}$ (this is the orange dot in Figure 1.3). This is your estimate of $y^{(1)}$. In

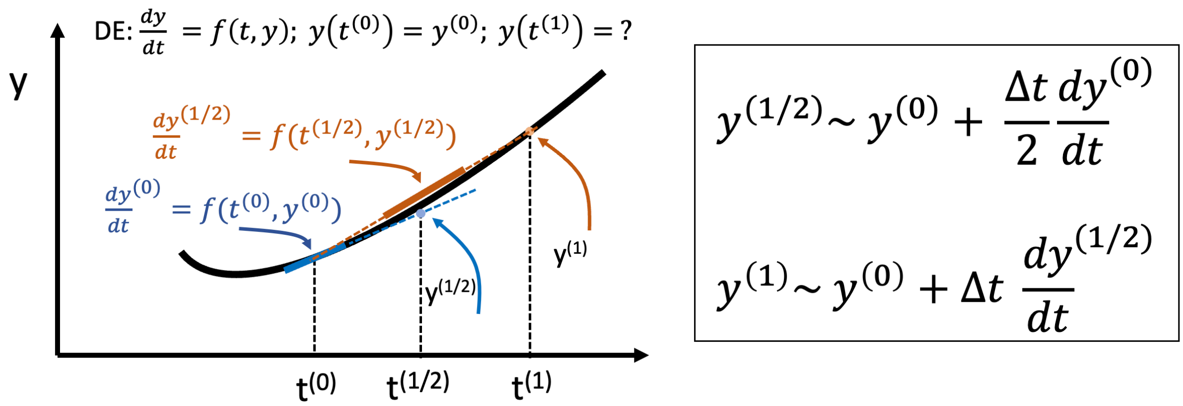

Figure 1.3: The midpoint method for solving a differential equation. For a walkthrough of the procedure see the text.

summary, you first calculate the derivative at the start of the time step and use it to estimate the the derivative at the midpoint (in time) then use that to advance the solution from the starting time to the end time. This is better than using the derivative at $t^{(0)}$, which would give you an answer where the blue line dashed crossed $t^{(1)}$. It is also better than using the midpoint evaluated derivative and advancing another half time step from $t^{(1/2)}$ to $t^{(1)}$.

1.5.2 Second order in space

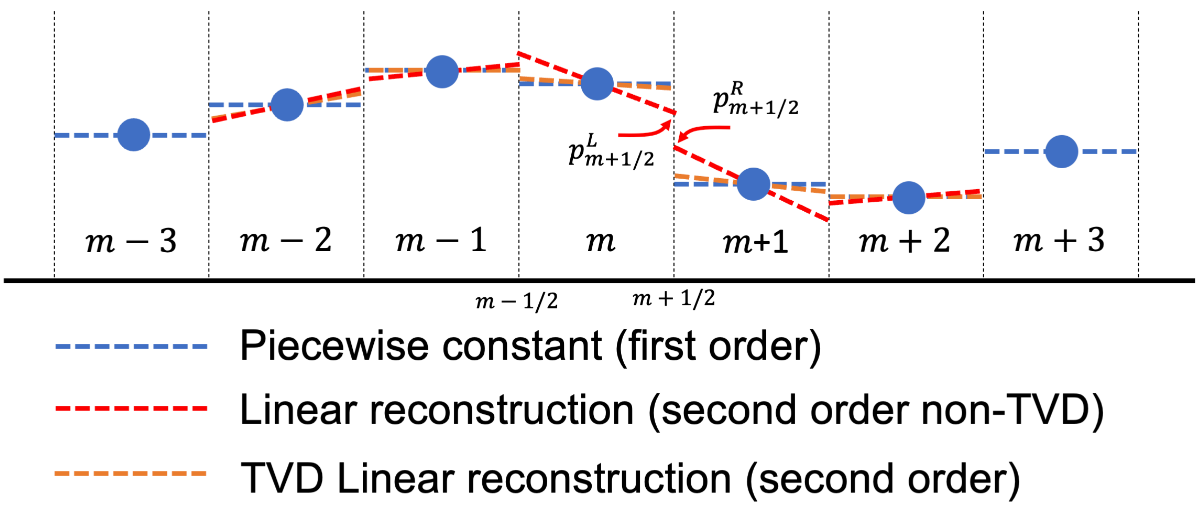

Figure 1.4: Examples of various first and second order reconstruction methods.

We when did the upwinding scheme in the last section we assumed that the density of the fluid passing through the interface was just the cell average value of the density. But, since we have densities in other zones, it is tempting to interpolate the density to the cell interface and use a better estimate of the density. This procedure is indeed often used and this is collectively known as *reconstruction*. One restriction we must follow is that once we reconstruct a density profile within the zone, it still must give the cell average value. We used piecewise constant reconstruction above, denoted via the blue lines in Figure 1.4. Furthermore, once we decide on the value of the flux at the interface, we must consistently use it in both numerical zones so that we conserve the the conserved variable.

The next obviously step is linear reconstruction. Basic linear reconstruction says the slope of the field within the zone is just set by the neighbouring zones; e.g. $\rho(x) = \rho_m + (x - x_m)\frac{\rho_{m+1}-\rho_{m-1}}{x_{m+1}-x_{m-1}}$. This can then be evaluated at $x = x_{m\pm1/2}$ for a estimation of the interface value. This is shown via the red lines in Figure 1.4. While it is a second-order method, it doesn't always give nice results. Extrema in the solution can develop at sharp interfaces. You can perhaps see why this might occur by looking at the red line in zone m or $m+1$ in Figure 1.4. The reconstructed value at the interface is larger than (or smaller than for $m+1$) the value in either zone. You can try this method in Exercise 2 below by using initial conditions that are not exactly smooth. Often in astrophysics we have shocks, i.e. jumps in pressure, density, energy and velocity, or other sharp features. A consistent numerical treatment of these is needed. To solve this, so-called TVD methods have been developed. TVD, or total variation diminishing, methods ensure than the total variation (between adjacent zones) in the solution at a later time is less than it is initially (i.e. it is diminished). If a scheme is a TVD scheme then extrema cannot develop. We can make our linear reconstruction scheme a TVD scheme by applying a limiter to the linear slopes we determine. The *minmod* limiter determines two slopes: $(\rho_{m+1} - \rho_m)/\Delta x$ and $(\rho_m - \rho_{m-1})/\Delta x$. If the slopes differ in sign, then we use piecewise reconstruction. If they have the same sign, the smaller (in absolute value) of the slopes is used. You can see the result of this with the orange lines in Figure 1.4. Since the slopes have different signs in zone $m-1$, and $m+2$, the orange line aligns with the blue. This method ensures that the reconstructed interface values never exceed any extreme of the zone or its neighbours.

A final note. For upwinding schemes the way forward is clear, we choose either the left side or right side reconstructed value based on the direction of fluid flow. However, for the Euler equations we need to be more careful. In terms of the Riemann problem mentioned above, we now formulate this as $[\mathbf{F}_i(\mathbf{U})]_{m\pm1/2} = \mathcal{R}(p^L_{m\pm1/2}, p^R_{m\pm1/2})$ where $p^{L/R}_{m\pm1/2}$ refer to the set of reconstructed primitive variables on the left (L) and right (R) side of the interface $m \pm 1/2$. A graphical depiction of $p^{L/R}_{m+1/2}$ is shown in Figure 1.4.

1.5.3 Exercise 2:

Repeat exercise 1, but with the new methods we developed here, show second order convergence of the solution. Note, typically one of the sources of error (either the time stepping or the spatial reconstruction) will dominate the error. Therefore you may increase the time stepping order and see no impact on the overall error. This means it is dominated by the spatial discretization. Also, while the TVD linear interpolation method we use here is nominally second order, when the minmod limiter reduces the interpolation to piecewise near extrema, the order of the method is reduced to first order. In fact, any so-called 'high resolution shock capturing' method reduces the order of the scheme at shocks to first order. You need smooth flows to achieve convergence at the theoretical order of the scheme.

This is maybe a good time to consider *guard cells*. Guard cells are extra cells in your domain, or more precisely directly outside it, that allow you to perform calculations on the main domain without having to worry about trying to access data in zones that do not exist. Without guard cells this becomes quite cumbersome for higher order methods where you may need data from several zones away from your own. You use boundary condition (like the periodic ones we used in exercise 1) to set the values of the guard zones anytime after the main domain has be updated.

1.6 Numerically solving the Euler equations

We are very close to being able to solve the Euler equations of Section 1.3. Let us first write down a discretized version of the one dimensional equations. We will keep it first order in time for now, Section 1.5.1 presents the methods needed to upgrade to second order in time, which is essentially just solving two equations of this form.

$$\frac{\mathbf{U}_m^{(n+1)} - \mathbf{U}_m^{(n)}}{\Delta t} = -\frac{[\mathbf{F}_x(\mathbf{U})]_{m+1/2}^{(n)} - [\mathbf{F}_x(\mathbf{U})]_{m-1/2}^{(n)}}{\Delta x} + \mathbf{S}(\mathbf{U}_m^{(n)})\,. \tag{1.12}$$

For the term, $[\mathbf{F}_x(\mathbf{U})]_{m+1/2}^{(n)}$, we need to solve the Riemann problem, which we have alluded to several times now, and state again for clarity:

$$[\mathbf{F}_x(\mathbf{U})]_{m\pm1/2}^{(n)} = \mathcal{R}(p_{m\pm1/2}^L, p_{m\pm1/2}^R)\,. \tag{1.13}$$

We have already discussed some methods of determining $p_{m\pm1/2}^{L/R}$ in Section 1.5.2. We also have used some trivial solutions to the Riemann problem for the advection problem (i.e. upwinding). But this isn't good enough for the Euler equations because not all of the equations can be written down like an advection equation. Even if they could, we also must consider that the fluid velocity isn't constant. For example we dropped the $\rho\partial_x u$ term in the expanded continuity equation in Section 1.4 because we assumed u was constant in space and time. In this short time for the project I think it is best we do not dwell to much on understanding the Riemann problem and its solutions. Although once we develop the methods I can come back to the problem hopefully use the simulations to enlighten you a bit about what solutions to the Riemann problem try to capture. In theory, for polytropic equations of state, the Riemann problem is exactly solvable using the Rankine–Hugoniot jump conditions, which are a representation of the Euler equations (i.e. conservation laws) for a discontinuous jump in the values of the primitives (typically called a shock). This is covered in AS7002. Take this approximation below, known as the HLLE Riemann solution, as your $\mathcal{R}$,

$$[\mathbf{F}_x(\mathbf{U})]_{m+1/2}^{(n)} \approx \frac{\lambda_R \mathbf{F}_x(\mathbf{U})(p_{m+1/2}^{L,n}) - \lambda_L \mathbf{F}_x(\mathbf{U})(p_{m+1/2}^{R,n}) + \lambda_R\lambda_L(\mathbf{U}(p_{m+1/2}^{R,n}) - \mathbf{U}(p_{m+1/2}^{L,n}))}{\lambda_R - \lambda_L}\,. \tag{1.14}$$

In this equation, $\mathbf{F}_x(\mathbf{U})(p_{m+1/2}^{L,n})$ is the flux term (i.e. in one dimension $\mathbf{F}_x(\mathbf{U}) = [\rho u, \rho u^2{+}P, (E{+}P)u]$) evaluated using the reconstructed primitives at the left side of the $m+1/2$ interface at the time slice (n) whereas $\mathbf{F}_x(\mathbf{U})(p_{m+1/2}^{R,n})$ is the same but at the right side of the interface (check out Figure 1.4 to see an example. Note how $p_{m+1/2}^{R,n}$ and $p_{m+1/2}^{L,n}$ are not necessarily the same, therefore $\mathbf{F}_x(\mathbf{U})(p_{m+1/2}^{R,n})$ and $\mathbf{F}_x(\mathbf{U})(p_{m+1/2}^{L,n})$ will not be the same. The Riemann solution is (in part) as weighted sum of these two different fluxes. $\mathbf{U}(p_{m+1/2}^{L,n})$ and $\mathbf{U}(p_{m+1/2}^{R,n})$ in Equation 1.14 are the conservative variables evaluated from the reconstructed primitives on the left and right sides of the interface.

The last bit is $\lambda_{R/L}$. These terms represent the wave speeds, or how fast information can travel (as sound waves or as a supersonic flow) in the domain. The weighted sum considers material on the left side of the interface moving the to right (i.e. $\lambda_R > 0$) and on the right side of the interface moving left (i.e. $\lambda_L < 0$). We can crudely estimate the wave speed by taking the $\lambda_R = \max(0, u_R + c_R^s, u_L + c_L^s)$ and $\lambda_L = \min(0, u_R - c_R^s, u_L - c_L^s)$. The 0s in here ensure that if the flow is moving supersonically in one direction (say $u_L \sim u_R > c^s$ than the $\lambda_L = 0$ and the interface flux reduces to $[\mathbf{F}_x(\mathbf{U})]_{m+1/2}^{(n)} = \mathbf{F}_x(\mathbf{U})(p_{m+1/2}^{L,n})$, as expected.

1.6.1 Exercise 3:

Now solve the system of Euler equations. To ensure you are capturing everything correctly it is best to start with a 'shock tube' problem. The classic shock tube initial data is the Sod problem. In this problem the left half of the domain, $x \leq 0.5$, the fluid is set of a density of 1 and a pressure of 1. In the right half of the domain, $x > 0.5$, the fluid is set to a density of 0.125

and the pressure is set to 0.1. The initial velocity everywhere is 0. The equation of state is taken as a polytrope with a $\gamma = 7/5$. Unlike the advection problem, the time step should consider not only the fluid speed but also the sound speed. Replace u in Equation 1.11 with $\max(c^s + |u|)$. Do not use periodic boundary conditions like we did for Exercise 1 and 2, rather, use reflecting boundary conditions. Think of this as a mirror placed at the domain edges, scalar quantities are just copied and velocity components perpendicular to the edge are flipped. For example, if the domain edge is at zone with an index of m and we want to use reflecting boundary conditions for the zones smaller then m, the density of the $m-1$ zone will have the value of the density in the m zone while the density of the $m-2$ zone will have the value of the density in the $m+1$ zone. For the velocity, the value of the velocity in the $m-1$ zones will be negative the value in the m zone.

As you might guess via intuition, the high-pressure high-density left side of the domain will flow into the right side. However the fluid dynamics here are complex, which is why this makes a great test problem of hydrodynamic solver. The resulting dynamics have a *shock wave*[4] propagating from the initial discontinuity toward the right side. There is also a *contact discontinuity*[5], and a *rarefaction fan*[6]. For polytropic equations of state the solution to many shock tube problems is analytic. I've included the analytic solution at a time of $t = 0.2$ in the course plan and show in Figure 1.5 the results for various resolutions.

1.6.2 Postmortem of Shock tube results and Riemann Problem

There is a very close relationship between a shock tube evolution and the Riemann problem for the Euler equations. Consider a zoomed in view of the interface $m + 1/2$ in Figure 1.4. The values on the left and right side of the interface are discontinuous, just like the values on the left and right side of the domain in the shock tube problem. In essence, solving the Riemann problem is just solving tiny little shock tubes at every interface. It is even easier really since we only care about the value of the flux right at the interface, i.e. at $x = 0.5$ in the shock tube problem. The goal of approximate Riemann solvers is to quickly get an estimate for the flux at the interface. Exact Riemann solvers exist, but are often much more computationally expensive. For quick reference, the values of the flux ($[\mathbf{F}_x(\mathbf{U})]^{(n)}_{m+1/2}$) given by the HLLE Riemann solver used above for the Sod shock tube initial conditions are $\sim 0.517, \sim 0.55,$ and ~ 1.331. Where as the fluxes at $x = 0.5$ in the Sod shock tube are $\sim 0.395, \sim 0.670,$ and ~ 0.943. These are taken at $t = 0.2$, but the flux at $x = 0.5$ is stable for all $t > 0$. Note that such large jumps, essentially a shock, naturally stress approximate Riemann solutions. For reference, I show the exact fluxes as a function of x for the Sod shock tube as well as the numerical solution at $t = 0.2$ in Figure 1.6. Once again, for the Riemann solution we are interested in the value at $x = 0.5$.

1.7 Exercise 4: Eulerian Hydrodynamics Project, AS8003

For the main project in AS8003 I want you to develop a two dimensional Eulerian hydrodynamics solver and use it to simulate two cases of common instabilities that arise in astrophysics. Like the shock tube you developed in Exercise 3, these are classic test problems for hydrodynamic codes. By instability I mean a hydrodynamical scenario that is an equilibrium state, but is unstable if perturbed a particular way. Furthermore these instabilities occur in astrophysics in many places and are often key to the evolution. Instabilities can be the source of mixing, turbulence, and amplification or several key fluid properties. The two instabilities I want you to simulate are the

[4]A shock wave is a discontinuity in density, pressure and velocity. Shocks are also present when fluid is moving supersonically and interacts with fluid at rest. Shocks generate entropy.

[5]Unlike a shock, a contact discontinuity is only discontinuous in density, and continuous in pressure and velocity (at leas the normal component). Contact discontinuities do not generate entropy.

[6]As the information that something dynamic is occurring travels to the left the fluid responds and expands into the lower density regime until it enters the supersonic regime. This expansion causes a rarefaction fan

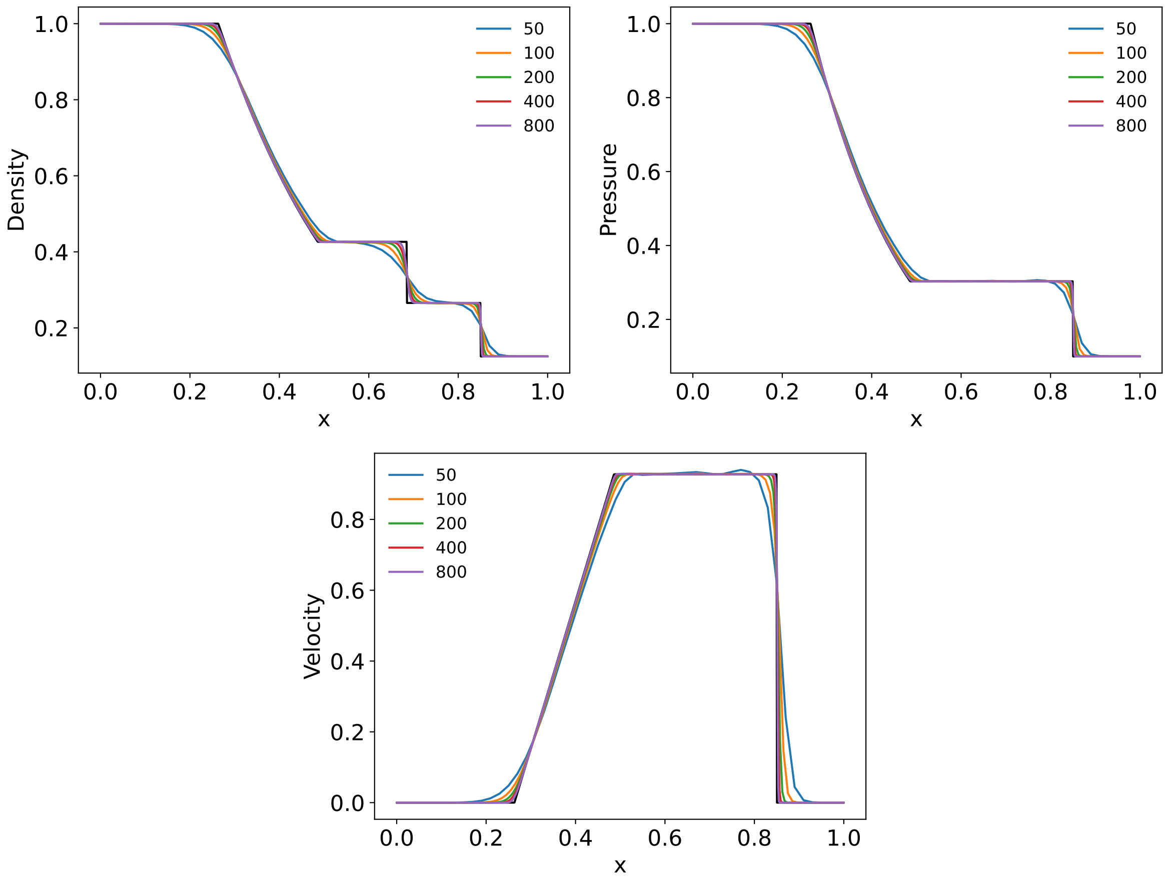

Figure 1.5: The solution of Exercise 3. Exact solution of the Sod shock tube problem (black) Also shown are numerical solutions of the Sod shock tube problem for domain sizes of 50, 100, 200, 400, and 800 at a $t = 0.2$. Shown are the density (top left), pressure (top right), and velocity (bottom).

Rayleigh-Taylor Instability (RTI) and the *Kelvin Helmholtz Instability* (KHI). Additional step now include implementing a fourth equation to solve (now ρu_y in addition to ρu_x is needed), adding the extra $\partial \mathbf{F_y}/\partial y$ flux term, and expanding the domain in a second dimension, the time step restriction typically goes as $\Delta t = C \min(\Delta x, \Delta y)/(\max(|u + c_s|) \times D)$ where D is the number of dimensions.

I have done exercise 4 in python with a solver that is probably not the most efficient, but that I spent a little bit of time on making efficient. For typical resolutions I'm asking you to look at, the runtime is not trivial. If you want to implement this solver in a compiled language I fully support that and can help you out if needed. You will likely see improvements in performance.

Part 1: RTI

The classic RTI we will consider is one where a dense fluid is placed next to a less dense fluid. Both fluids have the same adiabatic index. We impose an external gravitational field with a direction so the dense fluid is on the 'top'. The pressure in the fluid is chosen so that the fluid is in hydrostatic equilibrium (details below). This means the low density fluid just below the interface has a higher internal energy than the dense fluid just above the interface (recall $P = \rho\epsilon(\gamma - 1)$). If a low density fluid element is perturbed up, the pressure of the fluid element will change to be in equilibrium with the surroundings. However, the fluid element will change the pressure by changing the density, not the internal energy (since that takes some time to exchange heat with the surroundings). This means that as the pressure decreases the density decreases

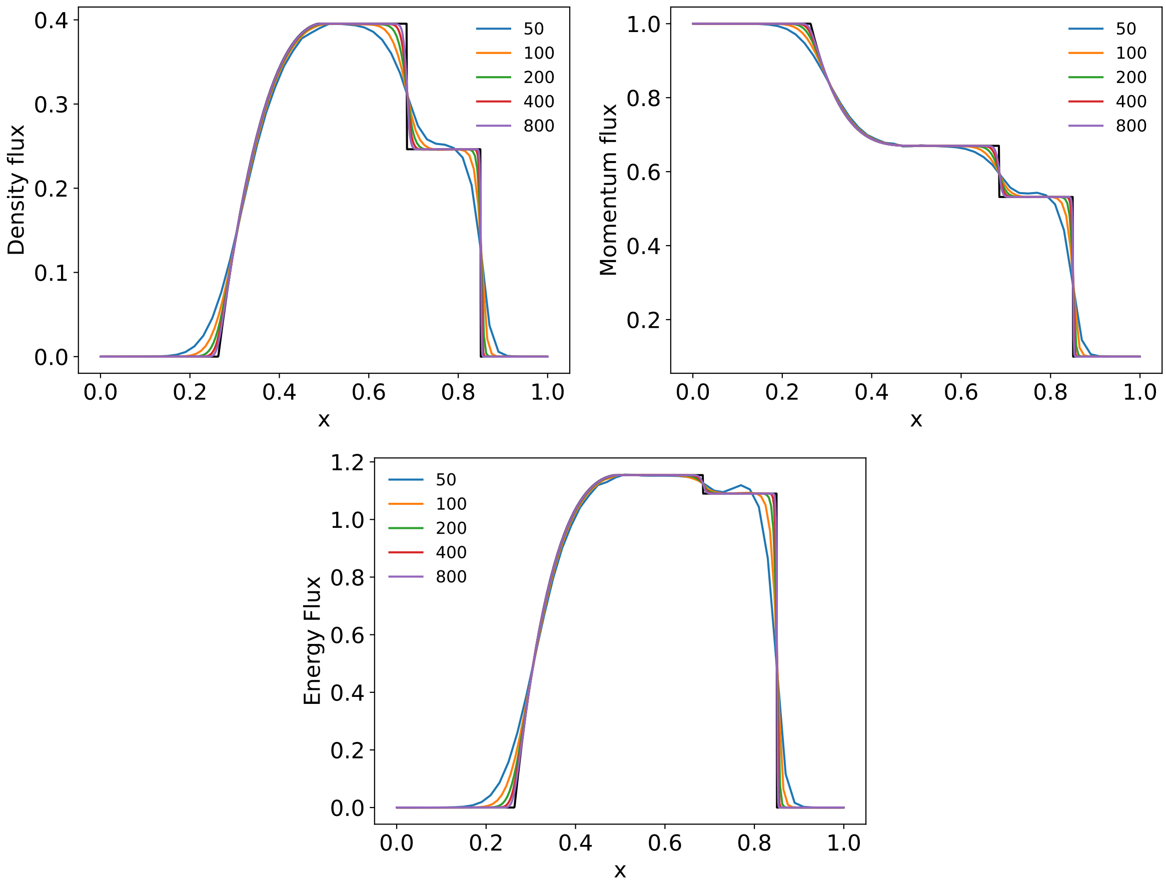

Figure 1.6: The solution of Exercise 3. Exact solution of the Sod shock tube problem (black) Also shown are numerical solutions of the Sod shock tube problem for domain sizes of 50, 100, 200, 400, and 800 at a time of 0.2. Shown are the density (top left), pressure (top right), and velocity (bottom).

according to the adiabatic index ($P \propto \rho^\gamma$). Since the internal energy of the perturbed element is higher than the surroundings, the density must be lower (since $P = \rho\epsilon(\gamma - 1)$ must equal the surroundings). The less dense fluid element will be buoyant in this new environment and will therefore accelerate up (against gravity). This is the key to the RTI. See a gas dynamics class for details of the derivation of the instability. A closely relate instability is the Richtmyer–Meshkov instability that occurs when a supernova shock wave traverses a star (See Figure 1.7 often the names are used interchangeably.

The initial conditions you should take are the following. They are based off of `https://www.astro.princeton.edu/~jstone/Athena/tests/rt/rt.html`, but common throughout the literature. Choose a box that has x dimensions of [-0.25:0.25] and y dimensions of [-0.75: 0.75] (i.e. three times taller than wide). Use a number of cells in the x direction of 100 and the number of cells in the y direction of 300. Use period boundary conditions in the x direction and reflecting in the y Depending on how efficient your code is (if you do this in a compiled language it shouldn't be a problem), feel free to increase the resolution to something like 200 × 600. Take a gravitational potential of $\phi = 0.1y$. This gives a uniform gravitational acceleration of $\vec{g} = -\vec{\nabla}\phi = -0.1\hat{y}$ (i.e. $g_y = -0.1$ in Equation 1.4). For $y < 0$, set $\rho = 1$ and for $y > 0$, set $\rho = 2$. To ensure hydrostatic equilibrium, $P = P_0 - \phi\rho = P_0 - 0.1y\rho$, take $P_0 = 2.5$. Use a polytropic equation of state with an adiabatic index of $\gamma = 7/5 = 1.4$. Once you set ρ and P, use the equation of state to set the internal energy via $P = \rho\epsilon_{\rm int}(\gamma - 1)$ and the sound speed via $c_s^2 = \gamma P/\rho$. Ideally use second order time integration and reconstruction. As described above,

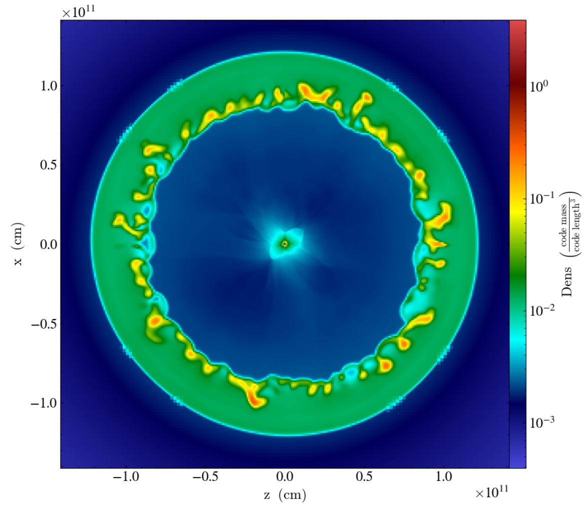

Figure 1.7: Example of a Rayleigh-Taylor like instability (precisely a Richtmyer–Meshkov instability) generated after a supernova shock wave passes through its parent star and an unstable reverse shock is formed.

we need to perturb the state to seed the instability. As discussed on that website, you can either perturb the density at the interface or the fluid velocity. It is better to perturb the velocities. Therefore set the y component of the velocity field to be $v_y = 0.01[1+\cos(4\pi x)][1+\cos(3\pi y)]/4$. Record the state of the fluid for a series of times while the instability develops, generating a movie of these states would be ideal for your presentation. The rough timescales are 5-15 time units. An snapshot of my results are shown in Figure 1.8. I use a second order in time evolution and both a second order (top) and a fifth-order (bottom) reconstruction. Both snapshots are shown at $t = 12.5$. Note that with the lower order (i.e. second order) scheme, the RTI is more diffusive. If during the week you can get everything working with the second order method, I can provide a function for WENO5 reconstruction, if you like.

Part 2: KHI

Another instability in the same class of instabilities is the Kelvin Helmholtz instability (KHI). This instability arises when two fluids have a shear layer between them, that is one layer of fluid is traveling with one speed while the other is traveling with a different speed. For the KHI, gravity is not required, however the presence of a gravitation field (and if the layers are RT-stable) was cause some modes (particularly small length scale modes) of the KHI to be stable. One very important astrophysical phenomena where the KHI is important is in neutron star-neutron star mergers. As the two neutrons approach each other in their orbit the surfaces begin to touch. At this point there is a large shear between the surfaces and they become KH-unstable. The instability causes the generation of turbulence, which on small scales can generation magnetic energy. This magnetic energy is likely crucial for generating the magnetic fields needed to launch jets and form a gamma ray burst in these systems (see Figure 1.9; from Palenzuela et al. (2022) and `https://prace-ri.eu/magnetic-field-amplification-in-binary-neutron-star-mergers/`).

The initial conditions you should take follow McNally et al. (2012). The domain extends for 0 to 1 in both the x and y dimension. Along the x direction (the direction of the flow) the boundaries need to be periodic, the y direction boundaries can be reflecting I've taken a 512×512 domain, but you can try with smaller one if the computations are too intense (i.e. 256×256). The adiabatic index is taken as $\gamma = 5/3$ and the pressure is everywhere taken as $P_0 = 2.5$. There is no gravitational potential $\phi = 0$ and therefore no gravitational acceleration.

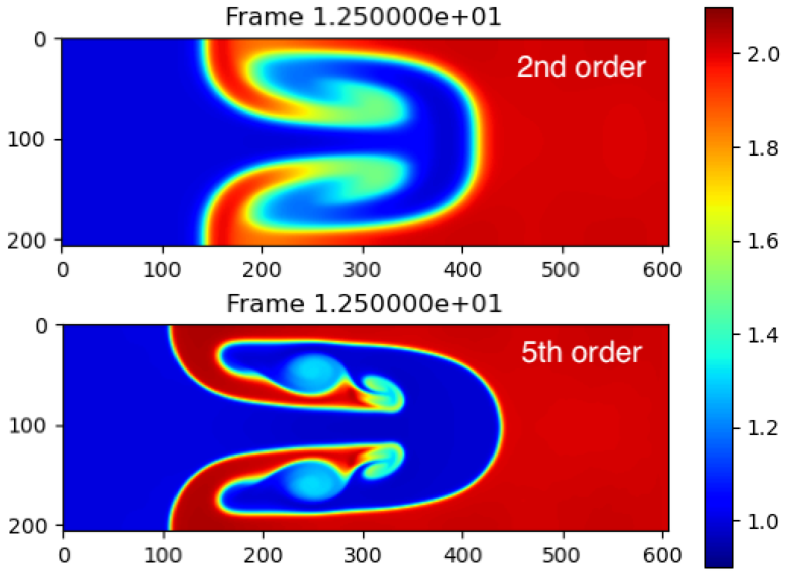

Figure 1.8: Final density configuration at 12.5 time units for the RTI setup described in the text. The top is using a second order reconstruction while the bottom uses a fifth order reconstruction. Both use the HLLE Riemann solver and second order time integration.

The initial density follows

$$\rho(x,y) = \begin{cases} \rho_1 - \rho_m \exp^{(y-1/4)/L} & \text{if } 0 \leq y \leq 1/4 \\ \rho_2 + \rho_m \exp^{(-y+1/4)/L} & \text{if } 1/4 \leq y \leq 1/2 \\ \rho_2 + \rho_m \exp^{-(3/4-y)/L} & \text{if } 1/2 \leq y \leq 3/4 \\ \rho_1 - \rho_m \exp^{-(y-3/4)/L} & \text{if } 3/4 \leq y \leq 1 \end{cases} \tag{1.15}$$

where $\rho_1 = 1$, $\rho_2 = 2$, $\rho_m = (\rho_1 - \rho_2)/2$, $L = 0.025$. The equation of state is then used to solve for the internal energy and speed of sound. The velocity in the x direction is as follows,

$$V_x(x,y) = \begin{cases} U_1 - U_m \exp^{(y-1/4)/L} & \text{if } 0 \leq y \leq 1/4 \\ U_2 + U_m \exp^{(-y+1/4)/L} & \text{if } 1/4 \leq y \leq 1/2 \\ U_2 + U_m \exp^{-(3/4-y)/L} & \text{if } 1/2 \leq y \leq 3/4 \\ U_1 - U_m \exp^{-(y-3/4)/L} & \text{if } 3/4 \leq y \leq 1 \end{cases} \tag{1.16}$$

where $U_1 = 0.5$, $U_2 = -0.5$, $U = (U_1 - U_2)/2$, $L = 0.025$. The equilibrium configuration is perturbed by setting a y velocity via

$$V_y(x,y) = 0.01 \sin(4\pi x)\,. \tag{1.17}$$

The rough timescales are 1-5 time units. A snapshot of my results are shown in Figure 1.10. I use a second order in time evolution and both a second order (left) and a fifth-order (right) reconstruction. Both snapshots are shown at $t = 3$. Note that with the lower order (i.e. second order) scheme, the KHI is more diffusive. As with the RTI, if during the week you can get everything working with the second order method, I can provide a function for WENO5 reconstruction, if you like. Advancing the instability further in time or using different initial perturbations are interesting directions you can take in your report.

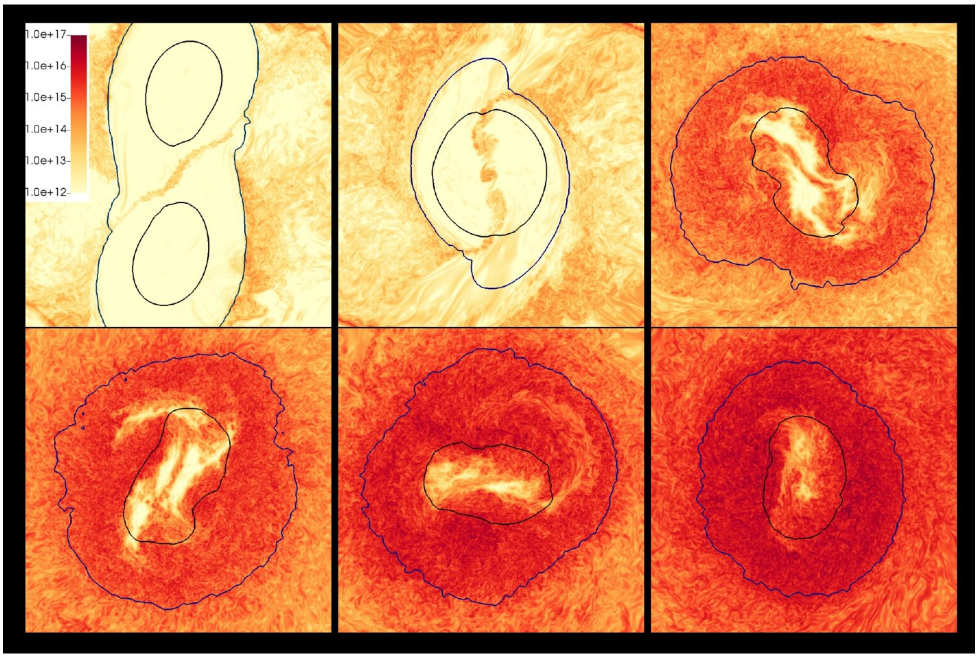

Figure 1.9: Example of the Kelvin Helmholtz instability at the interface between two merging neutron stars. The instability is particularly noticeable in the top middle panel. The color scale is the strength of the magnetic field (Palenzuela et al. 2022).

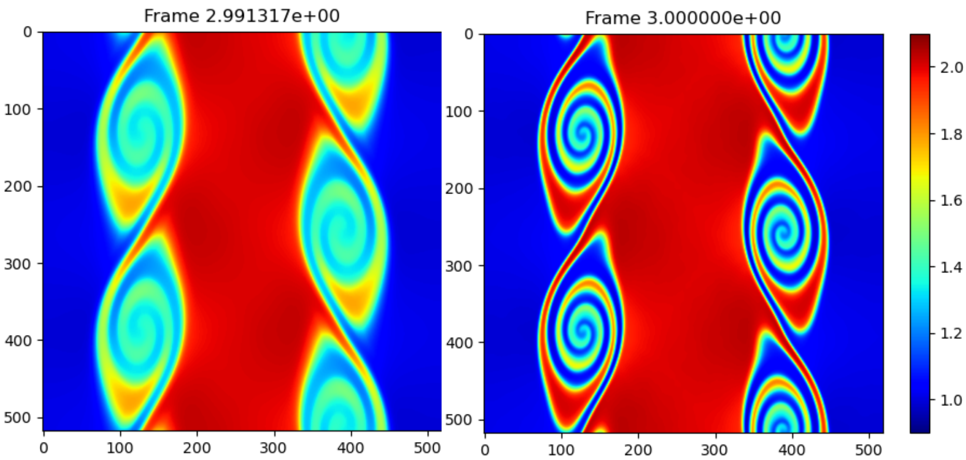

Figure 1.10: Final density configuration at 3 time units for the KHI setup described in the text. The left is using a second order reconstruction while the right uses a fifth order reconstruction. Both use the HLLE Riemann solver and second order time integration.

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